Section A

Cluster 1: Development, Economy and Environment

1 Resource 1 shows progress assessment for the 17 Sustainable Development Goals (SDGs) based on assessed targets. Resource 2A shows the proportion of national-level governments who partner with city-level governments to collect data on progress towards the Sustainable Development Goals in 2023. Resource 2B shows the proportion of countries with fully-funded strategies to collect data under implementation in 2023.

The Brahmaputra River is a transboundary river shared by China, India, and Bangladesh. Resource 3 shows the share of employment per economic sector for India from 2000 to 2022. Resource 4A is a map showing the Brahmaputra River and its riparian states. Resource 4B is an article on the proposed construction of the Yarlung Tsangpo Grand Dam on the Brahmaputra River.

(a) With reference to Resource 1, compare progress in SDG Goal 4 and Goal 8. [5]

Commented [ZY(1]: Goals 4 and 8 of the SDGs?

Award 1 mark for each comparison made on the progress in SDG Goal 4 and Goal 8, to a maximum of 5 marks.

Award a maximum of 1 additional mark for further development of each comparison, where applicable.

Possible responses include:

- Both SDG Goal 4 and Goal 8 show a similar overall level of progress. [1 mark]
  - Both Goal 4 and Goal 8 have combined "on track or target met", "moderate progress", and "marginal progress" segments each totalling approximately 50%. [1 additional mark]
- The degree of progress that is on track or has met its target is higher for Goal 4 than for Goal 8. [1 mark]
  - Goal 4 has 15% of progress that is on track or target met, while Goal 8 has 8% of progress that is on track or has met its target, a difference of approximately 7 percentage points. [1 additional mark]
- The degree of stagnation and regression is higher for Goal 8 than for Goal 4. [1 mark]
  - Goal 8 saw stagnation and regression totalling 32%, while Goal 4 saw stagnation and regression totalling 15%, a difference of 17 percentage points. [1 additional mark]

(b) Using Resources 2A and 2B, explain why there might be variations in data collection capacity across countries when measuring progress towards sustainable development. [4]

Award 1 mark for each reason explained, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each reason, where applicable.

Possible responses include:

- Variations in data collection capacity may arise due to differences in the ability of countries to collect data from cities within the country. [1 mark]
    - As shown in Resource 2, while 49% of countries have a formal process or established position for city-level data collection, 24% have rarely or no arrangement at all. Countries with formal processes would be better able to collect the city-level data needed to accurately measure progress towards sustainable development for the entire country, while countries with no arrangements would lack access to such data entirely, resulting in significant gaps in national-level data. [1 additional mark]

- Variations in data collection capacity may also arise due to differences in funding available for data collection across countries of different income levels. [1 mark]
    - As shown in Resource 3, 89% of high-income countries have adequate funding to collect data, compared to only 36% of low and lower-middle income countries. This large disparity of 53% suggests that lower income countries are significantly less equipped to collect the data needed to measure progress towards sustainable development, given that significant funding is needed to use advanced technologies, build data collection infrastructure, and hire skilled surveyors for data collection. [1 additional mark]

(c) Explain the political challenges faced by developed countries that limit progress towards sustainable development. [4]

Award 1 mark for each challenge explained, to a maximum of 4 marks.

Award a maximum of 1 additional mark for further development of each challenge, where applicable.

Possible responses include:

- A key political challenge is the lack of tangibility and immediacy of the positive impacts of sustainable development policies, making it difficult for political leaders in developed countries to justify the necessary trade-offs to their citizens. [1 mark]
    - In developed countries, where fossil fuels remain deeply embedded in the energy mix, political leaders face significant resistance when attempting to shift towards renewable energy. While the mitigation of climate change requires reducing carbon emissions in the present, the long-term benefits of doing so are difficult to communicate to citizens who are more concerned with immediate economic costs, such as higher energy prices and job losses in traditional industries. [1 additional mark]

- The uncertainty surrounding issues that limit progress towards sustainable development poses a significant political challenge, as it makes it difficult for political leaders to commit to concrete strategies. [1 mark]
    - In some developed countries where climate skepticism has gained political traction, the small degree of scientific uncertainty remaining around the causes and consequences of climate change continues to be exploited by those opposed to climate action. This makes it difficult for political leaders to build broad-based domestic support for a concrete sustainable development agenda. [1 additional mark]

- A lack of political will among governments in developed countries also limits progress towards sustainable development, as economic growth remains the primary pursuit of most governments. [1 mark]
    - In developed countries, where citizens might be sensitive to policies that may affect their economic wellbeing, governments frequently lack the will to implement measures such as carbon taxes or stricter environmental regulations that may be perceived as detrimental to economic development. This means that political leaders in developed countries often prioritise policies that deliver immediate economic returns over those that advance sustainable development. [1 additional mark]

- A lack of consensus on how to achieve sustainable development, and on the respective roles of different countries, poses a further political challenge that limits progress. [1 mark]

- Developed countries have at times prioritised their own self-interest over common global interests in international negotiations on sustainable development. For example, delays in the full ratification of the Kyoto Protocol were substantially due to the refusal of the United States to sign, on the basis that it would harm its domestic economy and that it unfairly favoured developing countries. [1 additional mark]

(d) Explain the limitations of Thomas Malthus' and Ester Boserup's views on the relationship between population growth and resources. [6]

Award 1 mark for each limitation explained, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each limitation, where applicable.

Possible responses include:

- Malthus underestimated human ingenuity/ intelligence and neglected the role of technological advancements in increasing food supply to support population growth, limiting the validity of his theory. [1 mark]
    - Improvements in scientific knowledge and agricultural innovation over time have undermined his assumption that food supply increases only in arithmetical progression. For example, the Green Revolution dramatically increased agricultural productivity in both developed and developing countries such as India, demonstrating that food supply can increase far faster than Malthus predicted. [1 additional mark]

- Boserup's perspective ignores the potentially limiting effect of environmental degradation on agricultural development, undermining her assumption that human ingenuity will always be able to increase food production with increasing population growth. [1 mark]
    - The Green Revolution itself illustrates this limitation, as intensive farming practices, heavy use of pesticides and fertilisers, and the burning of fossil fuels to power machinery have degraded soils needed for agricultural production, constraining rather than enabling further increases in food supply. [1 additional mark]

- A key limitation shared by both Malthus and Boserup is that both theories assume equal access to natural resources across all populations, when in reality access varies significantly due to differences in wealth between and within countries, which would mean that the inadequacy of quantity of resources is not due to population changes. [1 mark]
    - Generally, people in richer nations on a whole are wealthier and are therefore able to afford and access various types of resources, whether or not these resources are available in their geographical location naturally. In contrast, many people in poorer nations are less likely to be able to afford to purchase and consume resources, even if these resources might be available at or near their geographical region. [1 additional mark]

(e) Describe the changes in share of employment per economic sector for India from 2000 to 2022 shown in Resource 3. [5]

Award 1 mark for each change described, to a maximum of 5 marks.

Award a maximum of 1 additional mark for evidence provided for each change, where applicable.

Possible responses include:

- The primary sector saw an overall decrease in its share of employment from 2000 to 2022. [1 mark]
    - The share of employment in the primary sector decreased from 60% in 2002 to 45% in 2022, a decrease of 15%. [1 additional mark]

- The primary sector saw a decrease in its share of employment from 2000 to 2019, from 60% to 42%, before a slight increase to 45% by 2022. [1 mark]

- The secondary and the tertiary sectors saw an overall increase in their share of employment from 2000 to 2022. [1 mark]
    - The share of employment in the secondary sector increased from 15% in 2002 to 25% in 2022, an increase of 10%. The share of employment in the tertiary sector increased from 23% in 2002 to 30% in 2022, an increase of 7%. [1 additional mark]

(f) With reference to Resources 3 and 4, explain why there might be conflicts between riparian states on the Brahmaputra River Basin. [6]

Award 1 mark for each reason explained, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each reason, where applicable.

Response will be awarded a maximum of 3 marks if only either Resource 3 or Resource 4 is used.

Possible responses include:

- Conflicts may arise due to conflicting demands for water due to protection of self-interests that would impact the availability of water downstream as a result of the construction of the Yarlung Tsangpo Grand Dam. [1 mark]
    - China's proposed construction of the dam near the Great Bend is "central to its ambitions to achieve carbon neutrality by 2060 and stabilise its energy supply" and "projected to generate 60 gigawatts of power". This reflects China's prioritisation of its own economic and environmental goals in its use of water. [1 additional mark]

- However, the construction of the dam is likely to reduce the volume of water flowing downstream, creating conflicts with India and Bangladesh over the impact due to the dam's negative impact on their economic and social development. [1 mark]
    - The reduction in water flow would directly threaten agricultural productivity in both countries. The Brahmaputra currently supports "six million hectares of farmland in India's northeastern states" and "meets 55% of Bangladesh's irrigation needs". Furthermore, as sediment is trapped behind the dam, soil quality downstream would decline, further reducing agricultural productivity. [1 additional mark]

- The reduction in water availability would also threaten the livelihoods of a large proportion of the population in India and Bangladesh, given that the primary sector remains the largest source of employment in the region. [1 mark]
    - As shown in Resource 3, even in 2022, the primary sector still accounts for the largest share of employment in India at 45%. A reduction in water availability and agricultural productivity would therefore have a negative impact on the livelihoods of a significant proportion of the population, particularly farmer, heightening the potential for conflict between riparian states. [1 additional mark]

- These tensions are further compounded by the lack of a water-sharing treaty between China, India, and Bangladesh, as well as the broader context of declining rainfall and rising temperatures across the basin, which have already reduced total water availability in the region. [1 mark]

- Without a formal framework for managing shared water resources, there is no agreed mechanism for resolving disputes over water allocation, making it more difficult for riparian states to manage competing demands and increasing the likelihood of conflict as water scarcity intensifies due to climate change. [1 additional mark]

Cluster 2: Tropical Environments

1 In April 2024, Jiangwan Town, which lies in the Shaoguan City administrative area of Guangdong Province, China suffered extremely heavy rainfall, triggering mass movements and floods. Resource 5A shows the daily rainfall and cumulative rainfall distribution in Shaoguan in April 2024. Resource 5B shows the distribution of rainfall for the same period in April for the past 30 years in Shaoguan. Resource 6 shows a mass movement in Jiangwan Town on 24 April 2024.

Resource 7 shows the position of the Mississippi River delta lobes over different time periods.

(a) With reference to Resources 5A and 5B, explain why the rainfall events on 19 and 20 April 2024 in Shaoguan were considered "extreme". [6]

Award 1 mark for each explanation, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Responses must make reference to both resources to access full marks. Award a maximum of 3 marks for responses citing data from only one resource.

Possible responses:

From Resource 5A (Daily and Cumulative Rainfall, April 2024):

- The daily rainfall on 19 April 2024 was approximately 185 mm, which was exceptionally high compared to the rainfall on other days of the month [1 mark].
    - This single-day total was far greater than any other day in April 2024, with the next highest daily totals being around 100 mm recorded on 4 and 5 April [1 additional mark].

- The two-day rainfall event caused a sharp and sudden increase in the cumulative rainfall curve from 18-20 April [1 mark].
    - From 7 – 18 to April, cumulative rainfall had maintained at about 400mm. but this doubled to 800 mm after two consecutive days of 19–20 April. [1 additional mark]. **OR**
    - Over the two consecutive days of 19–20 April, Shaoguan received over 300 mm of rainfall which constituted more than a-third of April's rainfall over 2 days [1 additional mark].

From Resource 5B (30-year April Rainfall Distribution, 1992–2021):

- Compared to the 30-year historical record for April rainfall in Shaoguan, the total April 2024 rainfall of approximately 950 mm far exceeded any annual April total recorded between 1992 and 2021 [1 mark] **or** The rainfall on 19 April alone (~185 mm) already exceeded the highest recorded monthly April total in the past 30 years, demonstrating how anomalous the event was [1 mark].

- The highest April rainfall in the historical record appears to be approximately 125 mm (around 1996), making the 2024 event many times greater than the historical maximum [1 additional mark]. **OR**
- The historical data shows that April rainfall in Shaoguan typically ranges between approximately 20 mm and 125 mm [1 additional mark].

(b) Describe the type of mass movement shown in Resource 6. [4]

Award 1 mark for the correct identification of the type of mass movement shown in Resource 6. Accept rotational slide (slump), debris flow, or a combination of both as valid identifications, provided the candidate supports their answer with specific evidence from the photograph. Do not accept earthflow or translational slide as the primary identification.

Award 1 mark for each description of evidence from Resource 6 that supports the identification, to a maximum of 3 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

If the mass movement is not correctly identified, descriptions may still be credited for up to 3 marks, if they describe observable features from the photograph accurately.

Possible responses:

Identification:

- The mass movement shown is a **rotational slide (slump)** [1 mark], which may have subsequently transitioned into a debris flow at the base of the slope. [acceptable elaboration, not required for the identification mark]

Descriptions with supporting evidence from Resource 6:

- A **clearly defined, curved head scarp** is visible at the top of the slope where the material has detached from the hillside [1 mark].
    - This arcuate scar is characteristic of a rotational slide, where material moves along a concave, curved failure plane rather than a straight one [1 additional mark].

- The exposed slope face shows a large scar of bare, reddish-brown soil and regolith (weathered material), indicating that a significant mass of largely homogenous fine-grained material has moved downslope as a consolidated unit [1 mark].
    - This is consistent with a rotational slide, where near-homogenous material moves along a curved shear plane [1 additional mark].

- The debris has accumulated at the base of the slope in a spread-out mass, burying buildings, vehicles and infrastructure near the river in the foreground [1 mark].
    - The extensive spread and mixed nature of the accumulated material - including soil, vegetation and debris - is consistent with the toe of a rotational slide. [1 additional mark] **OR**
    - The mass movement may have transitioned into a debris flow at the base of the slope due to the accumulated water which may have saturated the slope material. [1 additional mark].

- The surrounding vegetation on either side of the scar remains largely intact, suggesting that the movement was confined to a specific mass along a defined failure plane [1 mark].
    - This is more consistent with a slide than a flow, where the entire slope surface would be more uniformly affected [1 additional mark].

(c) With reference to Resources 5 and 6, explain how the mass movement in Jiangwan Town of Shaoguan City may have occurred. [6]

Award 1 mark for each explanation of how the mass movement may have occurred, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses:

From Resource 5:

- The extreme rainfall on 19 and 20 April 2024, with daily totals of approximately 185 mm and 130 mm respectively, would have significantly **increased the water content of the slope materials** [1 mark].
    - The addition of water to the soil increases the weight of slope material, thereby increasing shear stress on the slope, making mass movement more likely [1 additional mark].

- The intense and prolonged rainfall recorded in Resource 5A over 19 and 20 April would have caused the soil and regolith on the forested slope in Resource 6 to become **saturated** [1 mark].
    - Saturation increases pore water pressure, which reduces friction between soil particles and lowers the shear strength of the slope material, making it more susceptible to failure [1 additional mark].

- The extreme rainfall recorded in Resource 5A on 19 and 20 April would have caused water to **infiltrate the forested slope** visible in Resource 6, **lubricating the curved failure plane** along which the rotational slide occurred [1 mark].
    - Water acting as a lubricant between the failure plane and the overlying material reduces friction, lowering shear strength and triggering the rotational movement visible in the photograph [1 additional mark].

- The large volume of water associated with the extreme rainfall in Resource 5A would have saturated the soil and regolith on the slope in Resource 6, causing the **toe of the rotational slide to become highly fluid** [1 mark].
    - This is consistent with the spread of mixed debris - including soil, vegetation, and structural material - visible at the base of the slope in Resource 6, suggesting that the toe of the rotational slide transitioned into a debris flow as water content increased downslope [1 additional mark].

(d) Describe how mass movement events may impact deposition in a river channel. [3]

Award 1 mark for each description of how mass movement events may impact deposition in a river channel, to a maximum of 3 marks.

Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses:

- Mass movement events introduce a **sudden and large influx of sediment load** into the river channel, leading to deposition both at the point of entry of the debris and/or further downstream [1 mark], where the river loses its ability to transport the increased load [1 additional mark].

- The input of **coarse material** from a mass movement event increases the bedload of the river, and this coarse material may remain deposited in the channel until discharge and velocity increase sufficiently [1 mark].

- Over time, repeated mass movement events can lead to **higher probability of deposition in general**, and hence the accelerated growth of mid-channel bars and/ or floodplains [1 mark].

(e) Explain how meanders may develop along a river. [7]

Award 1 mark for each explanation of how meanders may develop along a river, to a maximum of 6 marks.

Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses:

Conditions for meander formation:
- Meanders occur where vertical erosion has virtually ceased and slope angle is gentle, meaning the available energy is used in lateral erosion and deposition [1 mark].
    - Hence, meanders usually occur in the middle or lower course of a river where the gradient is gentler [1 additional mark].

Stage 1 — Alternate bar formation:
- Alternate bars form on opposite sides of the channel as a result of alternating zones of erosion and deposition [1 mark].
- The flow is diverted around these bars, converging as it moves towards the banks and diverging as it moves across the channels, which promotes erosion and pool formation on alternate sides of the channel [1 mark].

Stages 2 and 3 — Riffle formation:
- Deposition occurs at the crossing or inflection points where the flow diverges across the channel, forming riffles, which are shallow reaches within the channel [1 mark].
- At riffles, the velocity of the water is forced to reduce due to friction, and over time more sediment collects in the area, creating a point bar or slip-off slope [1 mark].

Stage 4 and 5 — Development of bends:
- The swing of the river flow induced by the riffles tends to direct the thalweg (zone of maximum velocity) to one side of the channel [1 mark].
    - This leads to erosion of the channel banks on the outer bank, producing a river cliff, while deposition continues on the inner bank, developing the meander bend further [1 additional mark].

Helicoidal flow and further meander development:
- The thalweg creates unequal pressure distribution — the water tends to "pile up" against the outer bank, resulting in a superelevation of water on that side of the channel [1 mark].
    - This leads to a compensatory flow of water across the channel from the outside (high pressure) to the inside (low pressure) of the bend, known as helicoidal flow — a corkscrew-like movement which spirals from one side of the channel to another between pools [1 additional mark].

- The helicoidal flow erodes the pools along the channel and increases deposition on the next inside bend after a pool [1 mark].
    - This further contributes to a point bar deposit on the inner (convex) bank and a deep pool near or at the outer (concave) bank, which then encourages the further formation and migration of meanders [1 additional mark].

(f) Describe the changes in the position of the delta lobes shown in Resource 7. [4]

Award 1 mark for each description of a change in the position of the delta, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses:

Changes in distance from the coastline:

- The delta lobes have progressively extended further from the coastline over time [1 mark].
    - The older lobes such as E and F are positioned closer to the current coastline and further inland, while the more recent lobes A and B extend further out into the Gulf of Mexico [1 additional mark].

Changes in direction of formation:

- The direction of delta formation has shifted from the north and west towards the south and east over time [1 mark].
    - The oldest lobe (F) is located furthest north near Lake Borgne, while the youngest lobe (A) is located furthest south and east, extending into the Gulf of Mexico [1 additional mark].

- The delta lobes show a general anticlockwise shift in direction of formation over time, moving from F in the north, through E and C in the west and south, to A in the southeast [1 mark].

Section B

Cluster 1 Development, Economy and Environment

2 Evaluate the potential of hydropower to reduce dependence on fossil fuels as an energy source. [20]

Possible Approaches:

Candidates could approach the question by making a judgment on the extent of hydropower's potential to reduce dependence on fossil fuels as an energy source through consideration of the strengths and limitations of hydropower in providing energy security, and the trade-offs associated with the use of hydropower that may impact its degree or extent of potential. Candidates could employ evaluative criteria such as spatial scale of implementation of hydropower, the temporal scale of the impacts, and the feasibility of utilisation of hydropower in various contexts.

Candidates could also approach the question by making a judgment on the extent of hydropower's potential to reduce dependence on fossil fuels as an energy source through a consideration of two or more case studies. Candidates could analyse the impacts in terms of contextual factors such as the geographical context and development level of different countries that might impact feasibility of adoption of hydropower.

Levels marked using Generic Level Descriptors for H2 essays.

3 Evaluate the view that global production networks (GPNs) of transnational corporations (TNCs) are of greater benefit to their host economies than their home economies. [20]

Possible Approaches:

Candidates could approach the question by making a judgment on whether the Global Production Networks (GPNs) of Transnational Corporations (TNCs) are of greater benefit to their host economies than their home economies through a consideration of criteria such as spatial scale of the impacts, the temporal scale of the impacts and the comprehensiveness of impacts. Candidates should consider a range of economic, social and environmental impacts, both positive and negative, to evaluate the significance of impacts.

Candidates could approach the question by making a judgement on whether the GPNs of TNCs are of greater benefit to their host economies than their home economies through a consideration of two or more case studies. Candidates could analyse the impacts in terms of contextual factors such as the development level of the economies, power relations between the state and TNCs, nature of TNC activities etc.

Levels marked using Generic Level Descriptors for H2 essays.

Cluster 2 Tropical Environments

4 Evaluate the impact of fluvial floods on places and people. [20]

Possible Approaches:

Candidates could approach the question by making a judgement on whether fluvial floods impact some places more than others through a consideration of criteria such as the comprehensiveness of the impact, spatial scale of the impact, and the temporal scale of the impact (long-term or short-term; delayed or immediate). Candidates could analyse the contextual factors such as the geographical/ locational context and socio-economic conditions which may have contributed to the variations in the impacts.

Candidates could also approach the question by making a judgement on whether fluvial floods impact some people more than others through a consideration of criteria such as comprehensiveness of the impact, spatial scale of the impact, and the temporal scale of the impact (long-term or short-term; delayed or immediate). Candidates could also consider the ability of the people to recover from the negative impacts. Candidates could consider different groups of people along the lines of gender, age and income level, etc.

Levels marked using Generic Level Descriptors for H2 essays.

5 Evaluate the view that geology is the determining factor influencing the formation of karst landforms in the humid tropics. [20]

Possible Approaches:

Candidates could approach the question by making a judgment on whether geology is the determining (i.e. main, or most important) factor through a consideration of criteria such as spatial scale of influence of the factor, temporal scale of influence of the factor, and whether the given factor is the root cause influencing formation of karst landforms in the humid tropics. Candidates should consider a range of factors in comparison to geological factors, such as climatic factors and human activities, to evaluate the significance of impacts.

Candidates could also approach the question by making a judgement on whether geology is the determining factor influencing karst landforms in the humid tropics through a consideration of two or more case studies. Candidates could analyse the impacts in terms of contextual factors such as geological and climatic factors relevant to the case studies raised.

Levels marked using Generic Level Descriptors for H2 essays.